

Food Chains Take-Home Assignment

Saturday, September 12, 2026 · 10:29 PM

homework

5th-6th

Science

60-90 min

AI as a thinking partner

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Redesigned for AI

Not AI-proof — designed so the thinking happens where you can see it.

AI USE · AI AS A THINKING PARTNER

Food chain concepts (definitions, energy flow, ecosystem impacts) are exactly the kind of content AI can generate fluently and generically, so the redesign needs students to anchor their answers in their own constructed example and reasoning, with AI usable for brainstorming or feedback rather than as the source of the content itself.

VULNERABLE STEPS

- Defining food chain, producer, consumer, decomposer
- Explaining how energy moves through an ecosystem
- Writing the paragraph on why the sun matters to food chains
- Creating a five-organism food chain example with explanations
- Writing the full 500-word essay on food chains and food webs

THINKING SAFEGUARDS

- Students must build their food chain around organisms from a specific, teacher-assigned or self-chosen local ecosystem, so AI can't supply a generic, accurate answer.
- A short in-class or conferenced check-in requires students to explain their food chain and defend what would happen if one organism disappeared, revealing whether they understand it or just transcribed it.
- Students submit a brief reflection disclosing any AI use and describing one thing they revised after getting feedback, making their process visible rather than just the final product.

Student AI Guidelines

Written for this task — ready to paste into the handout.

You may use AI as a thinking partner to brainstorm ideas, check your understanding, or get feedback on a draft — but you may not have it write your definitions, paragraphs, essay, or diagram explanations for you. Any AI use must be disclosed briefly at the end of your submission (what you asked it and how you used the response). Your food chain example must be based on a real ecosystem you choose or research yourself (not a generic textbook one), and you'll need to be ready to explain your food chain and answer a follow-up question about it in class or in a short conference. Show your own reasoning throughout — especially in the "what if one organism disappeared" question and your essay, where your own examples and word choices matter more than polished phrasing.

Food Chains Assignment

Name: _____

Due Date: _____

Part 1: Building Blocks (Draft stage — due first, checked before you continue)

1. In your own words, what is a food chain?
2. Explain the difference between a producer, consumer, and decomposer, using one example of each that you have not copied from a website.
3. Describe how energy moves through an ecosystem.
4. Write a paragraph explaining why the sun is important to almost every food chain.

Part 2: Your Ecosystem (choose one)

Pick one real ecosystem to focus on for the rest of this assignment — for example, a local pond, a forest near your home, a desert, or another specific ecosystem you're curious about. Name it and briefly describe it (2-3 sentences) before building your food chain.

5. Create a food chain with at least five organisms from your chosen ecosystem. Explain the role of each organism (producer, consumer, decomposer) and why it belongs at that point in the chain.
6. What might happen if one organism disappeared from your food chain? Explain the ripple effects on at least two other organisms, using your own reasoning.
7. Create a labeled diagram of your food chain.

Checkpoint: Before moving to Part 3, bring your Part 1 and Part 2 draft answers to class (or a short conference) to briefly walk through your food chain and answer one follow-up question about it.

Part 3: Essay (final piece)

8. Write a 500-word essay about the importance of food chains and food webs, using your own ecosystem example as at least one piece of supporting evidence. Include an introduction, three supporting paragraphs, and a conclusion.

Part 4: Reflection (required)

In 3-5 sentences, answer:

- Did you use AI at any point? If so, how (brainstorming, feedback, checking understanding)?
- What is one thing you changed or improved after getting feedback (from AI, a peer, or the teacher)?
- What part of this assignment made you think the hardest, and why?

Grading: Answers should be accurate, detailed, and reflect your own reasoning and choice of ecosystem. The checkpoint conversation and reflection are part of your grade, not just the final essay and diagram.

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The Original

What you started with, kept for comparison.

Food Chains Take-Home Assignment

Name: _____

Due Date: _____

Directions

Answer each question in complete sentences. You may use online resources. Type your responses and submit them electronically.

What is a food chain?

Explain the difference between a producer, consumer, and decomposer.

Describe how energy moves through an ecosystem.

Write a paragraph explaining why the sun is important to almost every food chain.

Create an example of a food chain with at least five organisms. Explain the role of each organism.

What might happen if one organism disappeared from your food chain?

Write a 500-word essay about the importance of food chains and food webs. Include an introduction, three supporting paragraphs, and a conclusion.

Create a labeled diagram of a food chain.

Grading: Answers should be accurate, detailed, and well written.

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The Conversation

What you asked, and what came back.

COACH

This is a solid foundation, and with a few tweaks we can keep it just as content-rich while making sure the thinking behind the answers stays visible — take a look below and let me know what you'd adjust.